Results

Time series research harness — Part 1: evaluation harness and baselines

Generated October 21, 2026

Contents

1 How this document works

This document reports the current results of the project, organised by topic: for each result, the question it answers, how it was tested, what came out, and what was concluded or decided from it. Design decisions themselves live in `DECISIONS.md` and `decisions/` (cited as Dn); open questions for supervision in `QUESTIONS.md` (Qn). How the document is kept is decided in D98.

Current results only. When a run is replaced (for example a rerun of the same configuration), its result leaves the text. Where a later result builds on it, it is kept as a sentence ("an earlier run gave ..."), not as an entry of its own.

Every run stays findable. All tables, figures and numbers quoted in the text are generated by `scripts/make_report.py` from the run folders (`runs/`, not committed); each entry names its run folders. The run index (Appendix A) lists every folder in `runs/`, including superseded, empty and not-yet-written-up runs, with what it was for, its status, commit and device.

Conventions. Unless stated otherwise: mean ± standard deviation over seeds (sample SD, $n-1$); the test set is scored once; accuracies are fractions, not percentages.

Reproduction criterion (D26 item 6, also used in D38). Implemented in `scripts/evaluate.py` and `scripts/tfc_official.py` as

$$|\text{published} - \text{mean}| \;\le\; \max(2\,\text{SD},\; 0.01).$$

Table 1: Reproduction checks: published accuracy against our mean over seeds, with the method's own probe and all labels. Criterion as implemented in `scripts/evaluate.py` and `scripts/tfc_official.py` (D26, D38): pass if $|\Delta| \le \max(2\,\text{SD}, 0.01)$.

Section	Method	Data	Published	Ours (mean ± SD)	$\lvert\Delta\rvert$	Allowed	Criterion
3.1	TS2Vec	FordA	0.9360	0.936 ± 0.003 ($n=3$)	0.000	0.010	pass
3.1	T-Loss	FordA	0.9280	0.932 ± 0.007 ($n=3$)	0.004	0.015	pass
3.1	MiniRocket	FordA	0.9500	0.946 ± 0.003 ($n=3$)	0.004	0.010	pass
3.1	TS-JEPA, official masking	FordA	0.9150	0.941 ± 0.005 ($n=3$)	0.026	0.010	fail
4.1	TF-C, official, full data	FD	0.8934	no test result	–	–	–
4.2	TF-C, ours	FD	0.8934	0.706 ± 0.112 ($n=5$)	0.187	0.224	pass
4.5	TF-C, official, subset	FD	0.8934	0.382 ± 0.164 ($n=5$)	0.511	0.327	fail

2 Data layer

2.1 Conversions reproduce the source values

Date 2026-10-07 – 2026-10-08
Decisions D3, D5, D20–D25, D27, D30, D39, D44
Runs none (unit tests; see below)

Question. Does the conversion to the canonical format (D3, D5) return exactly the values in the source files?

Test.

- Herning (`tests/test_herning.py`): the first 20 sweeps of `scope_20260901_112732.h5` are converted into a temporary folder; every sample of both channels (AE, UL) is compared with the source (absolute tolerance 10^{-9}, far below one code step), and a slice must equal the corresponding part of the full signal. The test also checks four tables, one telemetry row per capture, two channels and time order.
- FordA: converted values compared with the archive's text files.
- FD-A: checked for non-finite values before the TF-C divergence was investigated (D44).

Result. Herning conversion exact on the 20 sweeps tested; FordA exact against the files; FD-A contains no non-finite values. These outcomes are recorded in `HANDOFF.md` and D44; no test log is stored.

Finding. The data layer can be trusted for the runs below. The full Herning run has not yet been converted; the band check (§5.1) reads the source file directly.

Open: Test outcomes are not stored anywhere but in notes. Should test runs write a log (e.g. into `runs/`) so that this entry can cite one?

3 Public benchmark: FordA

FordA (UCR archive, used as distributed: D27, D30) is the first benchmark: frozen representations, the harness probes (D26), three seeds, label budgets 1 %, 10 % and 100 % (D29, D40). It validates the harness against published numbers and sets the bar the SSL methods have to clear against simple baselines (D75). The TS2Vec, T-Loss, MiniRocket and baseline runs were made on the work computer on 2026-10-20, one after another, with timing (D79). The working tree had uncommitted changes when they started (Table 9): the baseline code, committed afterwards in `a5b93a0`. TS-JEPA (2026-10-16) and PatchTST (2026-10-19; re-encoded with mean pooling in the cloud copy, D86, D87) were run earlier, before timing existed and also from uncommitted trees; they are used as they are, and rerunning them is deferred (D102).

3.1 TS2Vec, T-Loss and MiniRocket reproduce their published accuracies; TS-JEPA lands above its own

Date 2026-10-21
Decisions D26, D28, D32, D65, D66, D69, D76, D79, D81, D102, D103
Runs `20261020_144711_ts2vec_forda`, `20261020_144932_tloss_forda`, `20261020_152809_minirocket_forda`, `20261016_114600_tsjepa_forda`

Question. Do the methods, wrapped from their official code with the official settings, reproduce the published FordA accuracies with their own probes?

Test. TS2Vec with the official defaults (D28; 600 iterations, 320-dimensional representation max-pooled over time); T-Loss with its official settings (D32; 2,000 steps, $K = 10$); MiniRocket with the official code and settings (D76, D81; 9,996 features). Each is probed with its own protocol and all training labels: the RBF-SVM protocol of TS2Vec (also T-Loss's own), and MiniRocket's ridge classifier (D81). TS-JEPA from its official code with the official masking (D65, D66, D69; 5,000 epochs, learning rate $6 \cdot 10^{-5}$, mean over the 10 patch vectors), probed with the SVM, as TS2Vec and T-Loss: the paper's "small classification head" is not specified (D102, D103). Published targets: TS2Vec 0.936 (batch size 8), T-Loss 0.928, MiniRocket 0.950 (official results file, resample 0, taken to be the original split, D81), TS-JEPA 0.915 (paper, Table 1). Three seeds; TS2Vec, T-Loss and TS-JEPA on the GPU, MiniRocket on the CPU.

Result. TS2Vec 0.936 ± 0.003 (pass), T-Loss 0.932 ± 0.007 (pass), MiniRocket 0.946 ± 0.003 (pass), TS-JEPA 0.941 ± 0.005 (fail, above the published accuracy); Table 1. Earlier runs of the same TS2Vec and T-Loss configurations (2026-10-08, without timing; run index) gave 0.932 and 0.933: a rerun moves the mean by a few thousandths.

Finding. TS2Vec, T-Loss and MiniRocket reproduce, so these wrapped methods and the evaluation pipeline can be trusted on FordA. TS-JEPA does not reproduce, but it misses upwards: our accuracy is above the published one, not below. The paper's protocol differs from ours: its head is unspecified, and its TS2Vec number on FordA (0.864) is far below TS2Vec's own (0.936) and ours (D65). So the check cannot confirm the wrapping, but gives no sign that it loses accuracy.

3.2 SSL methods against training-free baselines: only TS-JEPA (masking fixed) keeps up at every budget

Date	2026-10-21
Decisions	D66, D69, D73–D80, D83, D85–D87, D89, D90, D102, D103
Runs	`20261020_143928_raw_forda`, `20261020_144122_fft_forda`, `20261020_144235_catch22_forda`, `20261020_152809_minirocket_forda`, `20261020_144711_ts2vec_forda`, `20261020_144932_tloss_forda`, `20261016_114616_tsjepa_forda_fixed`, `20261019_154248_patchtst_forda_mean`

Question. Do the SSL methods (TS2Vec, T-Loss, TS-JEPA, PatchTST) beat simple baselines on FordA, in accuracy, with few labels, and in speed (D75)?

Test. Baselines (D75): the floor (most frequent class, D83); the raw samples, probed with the SVM only (D89); the one-sided linear |FFT| spectrum (D76); catch24, i.e. catch22 plus mean and SD, from pycatch22 1.0.2 (D85, D90); and MiniRocket (D76, D81). Every baseline goes through the same probes, label budgets and seeds (0, 1, 2) as the SSL methods, so at each budget all methods are probed on the same labelled series (D78). Comparison: per probe and budget, each SSL method's paired difference to the best baseline (D80). TS-JEPA in its masking-fixed version only (D102: the official code takes every series of a batch from the first one); PatchTST with the mean over its patch vectors (D87), a choice made after seeing FordA test results (D86), which can only make PatchTST stronger. Timing: wall-clock per stage (D79), median over seeds; TS-JEPA and PatchTST not timed (D103).

Result. Accuracies in Table 2 and Figure 1; paired differences in Table 3; times in Table 4.

- TS-JEPA (masking fixed) is level with the best baseline at 10 % and 100 % labels (logistic −0.001 and +0.002, SVM −0.001), except with the SVM at 10 %, where it is ahead by +0.010 in every seed.
- TS2Vec and T-Loss at 10 % and 100 % labels: the best baseline is ahead with both probes, in every seed. With the logistic probe at 10 %, |FFT| reaches 0.933 against 0.904 for TS2Vec (paired difference −0.029; T-Loss −0.021).
- 1 % labels: with the SVM probe, TS2Vec, T-Loss and TS-JEPA are ahead of |FFT| by +0.028, +0.023 and +0.024, largely from seed 0. With the logistic probe TS2Vec and T-Loss tie with MiniRocket (+0.002 and −0.007); TS-JEPA is ahead by +0.012, again mostly from one seed.
- PatchTST is below the best baseline at every budget and in every seed, by −0.071 (SVM, 100 %) to −0.248 (logistic, 1 %).
- Speed: per seed, MiniRocket fits in 0.08 s on the CPU, TS2Vec trains in 24.20 s and T-Loss in 12.4 min on the GPU. |FFT| and catch24 fit nothing. The slowest step for MiniRocket is the SVM probe on its 9,996 features. TS-JEPA and PatchTST are not timed.
- catch24 is the weakest baseline after the raw samples, and its spread at 1 % labels is large.

Finding. On FordA, TS-JEPA with the masking fixed is the only SSL method that keeps up with the training-free baselines at every budget. It is ahead only at 1 % labels (mostly one seed) and with the SVM at 10 % (+0.010). TS2Vec and T-Loss pay off only with 1 % labels and the SVM probe, and not in every seed; PatchTST is well below the baselines throughout. Whether TS-JEPA's small margin is worth its training cost is open until it is timed; the baselines need no training (MiniRocket fits in 0.08 s). FordA's classes are well separated by the spectrum, so this need not carry over to the rig data; it sets the bar: an SSL method has to beat |FFT| and MiniRocket, not only the floor.

Open: TS-JEPA and PatchTST have no timing; rerunning them with timing is deferred (D102). The untrained-encoder control (D75 f) has not been run. On the GPU the first seed includes start-up time (TS2Vec: 30 s against 23–24 s for the others); the table reports the median.

Table 2: FordA test accuracy, frozen representations: mean ± SD over 3 seeds; the same labelled series for every method at each budget (D78). Bold: best mean per column. Raw samples are not probed with logistic regression (D89). The floor ignores the features, so it is the same under both probes (D83).

	Logistic regression			SVM (TS2Vec protocol)		
Method	1 %	10 %	100 %	1 %	10 %	100 %
TS2Vec	0.862 ± 0.022	0.904 ± 0.011	0.929 ± 0.004	**0.876** ± 0.025	0.914 ± 0.006	0.936 ± 0.003
T-Loss	0.853 ± 0.018	0.912 ± 0.006	0.928 ± 0.004	0.871 ± 0.018	0.911 ± 0.007	0.932 ± 0.007
TS-JEPA (masking fixed)	**0.871** ± 0.018	0.932 ± 0.005	**0.947** ± 0.008	0.873 ± 0.020	**0.936** ± 0.004	0.945 ± 0.004
PatchTST	0.612 ± 0.040	0.794 ± 0.057	0.855 ± 0.045	0.689 ± 0.022	0.798 ± 0.042	0.875 ± 0.030
Raw samples	not run	not run	not run	0.558 ± 0.035	0.669 ± 0.008	0.828 ± 0.000
|FFT|	0.829 ± 0.011	**0.933** ± 0.004	0.939 ± 0.000	0.848 ± 0.018	0.925 ± 0.005	0.945 ± 0.000
catch24	0.664 ± 0.129	0.830 ± 0.010	0.843 ± 0.000	0.602 ± 0.104	0.838 ± 0.004	0.880 ± 0.000
MiniRocket	0.860 ± 0.032	0.923 ± 0.007	0.945 ± 0.004	0.795 ± 0.041	0.917 ± 0.003	**0.946** ± 0.001
Floor (most frequent class)	0.516	0.516	0.516	0.516	0.516	0.516

Table 3: FordA: paired difference in test accuracy, SSL method minus the best baseline at that probe and label budget (positive: the SSL method is better). Per seed, the two use the same labelled series (D78). The best baseline is chosen by its mean test accuracy, which favours the baselines (D80). Seeds in the order 0, 1, 2.

Probe	Labels	Best baseline (mean)	TS2Vec: mean (per seed)	T-Loss: mean (per seed)	TS-JEPA (masking fixed): mean (per seed)	PatchTST: mean (per seed)
Logistic regression	1 %	MiniRocket (0.860)	+0.002 (−0.013, +0.011, +0.009)	−0.007 (−0.021, −0.022, +0.022)	+0.012 (−0.001, +0.003, +0.033)	−0.248 (−0.259, −0.292, −0.192)
	10 %	|FFT| (0.933)	−0.029 (−0.035, −0.036, −0.016)	−0.021 (−0.023, −0.024, −0.015)	−0.001 (−0.011, +0.004, +0.005)	−0.139 (−0.080, −0.152, −0.185)
	100 %	MiniRocket (0.945)	−0.017 (−0.018, −0.019, −0.014)	−0.018 (−0.010, −0.027, −0.017)	+0.002 (+0.007, −0.011, +0.011)	−0.091 (−0.040, −0.100, −0.133)
SVM (TS2Vec protocol)	1 %	|FFT| (0.848)	+0.028 (+0.067, +0.002, +0.014)	+0.023 (+0.055, −0.014, +0.029)	+0.024 (+0.060, −0.002, +0.016)	−0.160 (−0.146, −0.203, −0.130)
	10 %	|FFT| (0.925)	−0.011 (−0.003, −0.019, −0.011)	−0.014 (−0.010, −0.020, −0.011)	+0.010 (+0.015, +0.006, +0.010)	−0.128 (−0.077, −0.135, −0.171)
	100 %	MiniRocket (0.946)	−0.010 (−0.009, −0.013, −0.009)	−0.014 (−0.013, −0.021, −0.008)	−0.001 (−0.002, −0.004, +0.003)	−0.071 (−0.039, −0.080, −0.096)

Table 4: FordA: wall-clock time per stage (D79), median over 3 seeds. Encoding: training and test series (4,921) per second. Probe fit includes the probe's cross-validation. Hardware: NVIDIA RTX 4500 Ada Generation GPU, 36-thread CPU. MiniRocket's code compiles once on import (0.54 s, not included, D81). TS-JEPA and PatchTST were run before timing existed (D102, D103).

Method	Device	Fit / pretraining (s) median (range)	Encoding series/s	Probe fit, 100 % labels (s) Logistic	SVM
TS2Vec	GPU	24.20 (22.58–29.56)	2,577	2.8	9.1
T-Loss	GPU	742.81 (738.37–756.01)	829	2.9	9.9
TS-JEPA (masking fixed)	–	not timed	–	–	–
PatchTST	–	not timed	–	–	–
Raw samples	CPU	none	639,095,705	–	30.4
|FFT|	CPU	none	294,389	9.5	7.3
catch24	CPU	none	1,098	0.7	73.3
MiniRocket	CPU	0.08 (0.07–0.10)	28,549	33.5	540.7

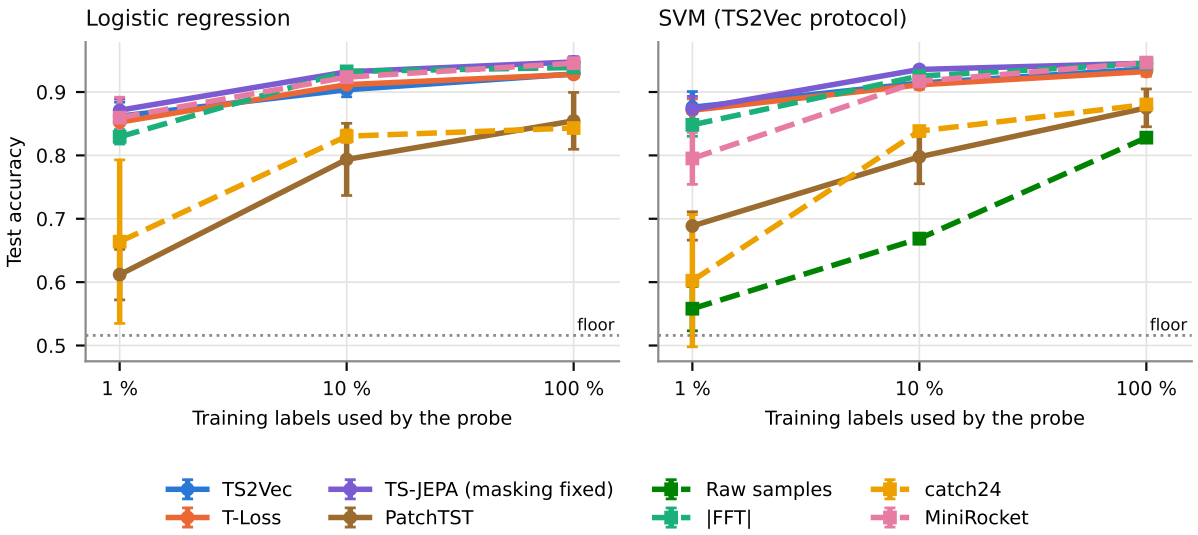


Figure 1: FordA test accuracy against the share of training labels given to the probe, one panel per probe. Solid lines with circles: SSL methods; dashed lines with squares: baselines; dotted: the floor. Error bars: SD over 3 seeds. Raw samples appear only under the SVM probe (D89). The numbers are in Table 2.

4 TF-C on FD-A → FD-B

TF-C is studied in steps (D35): (1) reproduce the published result with the official code; (2) the official TF-C on the full, raw Paderborn data; (3) our own corrected TF-C; (4) possibly a comparison on FordA. FD-A and FD-B are used as distributed by the TF-C authors (D39): pretraining on FD-A (train), fine-tuning on the 60 FD-B training windows, epoch selection on the 21 FD-B validation windows, testing on FD-B test. Target: 0.8934 ± 0.0379 accuracy, 5 runs (TF-C paper, Table 5; D38). The official reference is the 2023 code (D36), run with two protocols: *official* (epoch with the best *test* accuracy, as the official code) and *clean* (epoch with the best validation macro-F1, D37, D43).

4.1 Official TF-C diverges on the full FD-A data

Date 2026-10-08
Decisions D36, D37, D38, D44
Runs `20261008_134951_tfc_official_fd`, `20261008_143021_tfc_official_fd`, `20261008_155516_tfc_official_fd`

Question. Does the official TF-C code, run on all FD-A windows (debugging subset off, D37), reproduce the published FD-A → FD-B accuracy?

Test. Official 2023 code through our adapter, settings checked against the official `FD_A_Configs.py` (40 epochs, batch 64, learning rate $3 \cdot 10^{-4}$, weight decay $3 \cdot 10^{-4}$ as hard-coded). Five seeds. Three attempts: the first (seed 0 only) and second with commit `87e9222`; the third with commit `c11957a`, which records a non-finite pretraining loss as divergence and skips fine-tuning for that seed (D44).

Result. In 4 of 5 seeds the pretraining loss became non-finite (Table 5, Figure 4); in the first attempt fine-tuning then crashed. The pretraining losses of the second and third attempt are identical for every seed, and the first attempt's finite losses (seed 0) are identical to them, so the divergence is deterministic given the seed. Seed 4 completed pretraining, but no attempt wrote fine-tuning output for it, and the third attempt wrote no `results.csv` (results file written: no).

Finding. On the full data, the official code does not produce a single test result. A likely cause, not verified (D44): gradients are reset only once per epoch, so each step on the full data uses many accumulated batches.

Consequence. D44: divergence is now recorded per seed as a result, and the official run was repeated with the debugging subset (§4.5).

Open: Why seed 4, whose pretraining finished, has no fine-tuning output, is not recorded in the run folder.

4.2 Our TF-C falls well below the published accuracy

Date 2026-10-08
Decisions D41, D42 (D38, D39)
Runs `20261008_142751_tfc_fd`

Table 5: Official TF-C, full FD-A pretraining data (third attempt). Pretraining loss per epoch is empty in `pretrain_loss.csv` once it is no longer finite.

Seed	Epochs run	Finite	First non-finite	Last finite loss	Fine-tuning output
0	5	4	5	8.290	no
1	22	21	22	8.186	no
2	17	16	17	8.190	no
3	5	4	5	8.340	no
4	40	40	–	8.185	no

Second and third attempt: pretraining losses identical for every seed.

Question. Does our TF-C (paper's CNN and loss, corrected frequency augmentation, selection by validation macro-F1; D41, D42) reach the published FD-A → FD-B accuracy?

Test. Settings in D41/D42 and the run's `config.yaml`. Five seeds. Component checks in `tests/test_tfc.py` (NT-Xent and L_C equal the official functions; frequency augmentation symmetric and the spectrum of a real signal; permutation reaches M segments); their outcome is not stored with the run.

Result. Test accuracy 0.706 ± 0.112, macro-F1 0.757 ± 0.089; single seeds range from 0.519 to 0.803 (Table 6). The validation macro-F1 moves in large steps between epochs (Figure 2); one validation window is 1/21 of the validation set.

Finding. The mean is far below the published 0.8934, with a large spread over seeds. Formally, the criterion of D26 gives "pass", because twice the SD exceeds the gap (Table 1).

Consequence. D45 and D46: diagnostics to tell "pretraining learned little" from "fine-tuning or selection failed" (§4.3) and to check whether FD-B can separate methods (§4.4).

Open: The criterion passes a mean far below the target because the spread is large. Is our TF-C result a reproduction? Should the criterion get a condition on the spread (D26 item 6 speaks of the 1-pp clause only "if the spread is very small", while the code applies it always)?

Table 6: Our TF-C (D41, D42), FD-A → FD-B test set. Epoch: fine-tuning epoch with the best validation macro-F1; the test set is scored once, for that epoch. Precision, recall and F1 are macro averages.

Seed	Epoch	Acc.	Prec.	Recall	F1	AUROC	AUPRC
0	26	0.803	0.857	0.839	0.847	0.765	0.738
1	40	0.769	0.844	0.831	0.808	0.831	0.812
2	33	0.519	0.638	0.648	0.614	0.732	0.712
3	18	0.690	0.764	0.769	0.745	0.847	0.821
4	30	0.750	0.752	0.817	0.769	0.786	0.758
Mean ± SD		0.706 ± 0.112	0.771 ± 0.088	0.781 ± 0.079	0.757 ± 0.089	0.792 ± 0.047	0.768 ± 0.047

4.3 Frozen probes on our TF-C encoders: pretraining is not where the shortfall is

Date 2026-10-08
Decisions D45
Runs `20261008_151014_probe_tfc_fd`

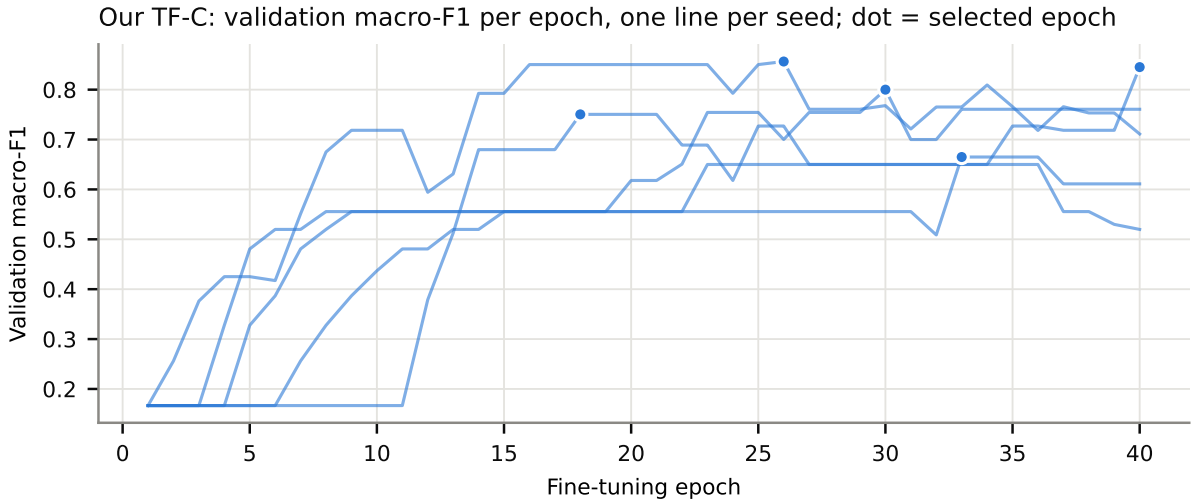


Figure 2: Our TF-C: validation macro-F1 after every fine-tuning epoch, one line per seed; dot: the selected epoch (first maximum).

Question. Did our TF-C pretraining (§4.2) learn a representation that separates the FD-B classes, so that the low fine-tuned accuracy comes from fine-tuning or epoch selection? Or did pretraining learn little?

Test. The five pretrained encoders of our TF-C run are frozen; their representation $[z_T; z_F]$ is probed with both harness probes trained on all 60 FD-B training labels and scored once on FD-B test (D45). Reference: untrained encoders of the same architecture, initialised with the same seeds, probed the same way. 100 % labels only (exception to D40: 1 % of 60 is less than one window).

Result. Pretrained encoder: SVM 0.951 ± 0.021, logistic regression 0.961 ± 0.012. Untrained encoder: SVM 0.973 ± 0.020, logistic regression 0.738 ± 0.091 (Table 8, Figure 3).

Finding. Probes on the frozen pretrained encoders score far above the fine-tuned TF-C (§4.2), which points to fine-tuning or epoch selection rather than pretraining. But pretraining does not beat an untrained encoder with the SVM probe; it helps only the linear probe.

4.4 Plain spectra separate FD-B almost perfectly

Date	2026-10-08
Decisions	D46
Runs	`20261008_162447_probe_spectrum_fd`

Question. Can FD-B separate methods at all, given that an untrained encoder with an SVM probe already reaches high accuracy (§4.3)?

Test. Both harness probes trained directly on the full FFT magnitude spectrum of each FD-B training window (the input of TF-C's frequency branch), no encoder; 60 labels; scored once on FD-B test. Deterministic, so one value per probe. Run in Claude's cloud workspace on the work computer's FD-B data, copied unchanged (`NOTE.txt` in the run folder).

Result. SVM 0.999, logistic regression 0.943 (Table 8).

Finding. With 60 labels, an SVM on the plain spectrum classifies FD-B test almost without error, above the published TF-C result. Under this protocol FD-B cannot show that a pretrained method adds anything (the condition stated in D46's rationale is met).

Open: What follows for the TF-C study (D35) from the FD-B diagnostics (§4.3–§4.5) has not been decided.

4.5 Official TF-C with its debugging subset collapses to one class

Date 2026-10-08
Decisions D37, D44 (D43)
Runs `20261008_171159_tfc_official_fd_subset`

Question. Is the debugging subset what makes the official code trainable, and does it then reproduce the published accuracy (D37: first check if the reproduction fails; D44)?

Test. As in §4.1, with the official debugging subset switched on (as shipped: 640 FD-A windows for pretraining). Five seeds; both protocols.

Result. No seed diverged (Figure 4). Test accuracy 0.382 ± 0.164, macro-F1 0.178 ± 0.069 (official protocol; criterion: fail). In every seed and every fine-tuning epoch, macro recall on test and validation is exactly 1/3 (Table 7), which is what predicting one class for every window gives. All epochs therefore tie and both protocols select epoch 1; official and clean numbers are identical.

Finding. With the subset the official code trains without diverging, but the fine-tuned model predicts a single class. Neither configuration of the official code reproduces the paper.

Table 7: Official TF-C with the debugging subset. Official protocol: epoch with the best test accuracy; clean protocol: epoch with the best validation macro-F1 (D43). Last column: macro recall on test and validation is exactly 1/3 at every one of the 40 fine-tuning epochs.

Seed	Official protocol			Clean protocol			Recall = 1/3
	Epoch	Acc.	F1	Epoch	Acc.	F1	all epochs
0	1	0.455	0.209	1	0.455	0.209	yes
1	1	0.089	0.055	1	0.089	0.055	yes
2	1	0.455	0.209	1	0.455	0.209	yes
3	1	0.455	0.209	1	0.455	0.209	yes
4	1	0.455	0.209	1	0.455	0.209	yes

FD-B results side by side

Table 8: FD-B test set: TF-C results and the diagnostics. Probes use all 60 FD-B training labels (D45, D46). The |FFT| probes are deterministic (one value).

Model / probe	Section	Accuracy	Macro-F1	Seeds
TF-C paper (published)	–	0.893 ± 0.038	0.916 ± 0.083	5
Official TF-C, subset, fine-tuned	4.5	0.382 ± 0.164	0.178 ± 0.069	5
Our TF-C, fine-tuned	4.2	0.706 ± 0.112	0.757 ± 0.089	5
SVM on pretrained TF-C encoder	4.3	0.951 ± 0.021	0.964 ± 0.015	5
log. reg. on pretrained TF-C encoder	4.3	0.961 ± 0.012	0.971 ± 0.009	5
SVM on untrained encoder	4.3	0.973 ± 0.020	0.980 ± 0.015	5
log. reg. on untrained encoder	4.3	0.738 ± 0.091	0.793 ± 0.085	5
SVM on |FFT| spectrum	4.4	0.999	0.999	1
log. reg. on |FFT| spectrum	4.4	0.943	0.958	1

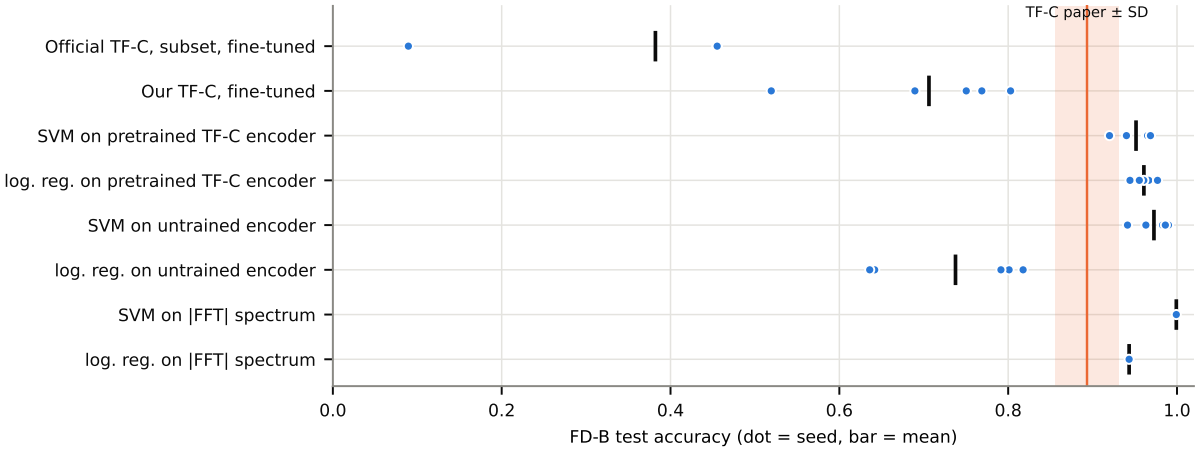


Figure 3: FD-B test accuracy for every TF-C model and probe: dots are seeds, bars means. Shaded: published TF-C accuracy ± SD.

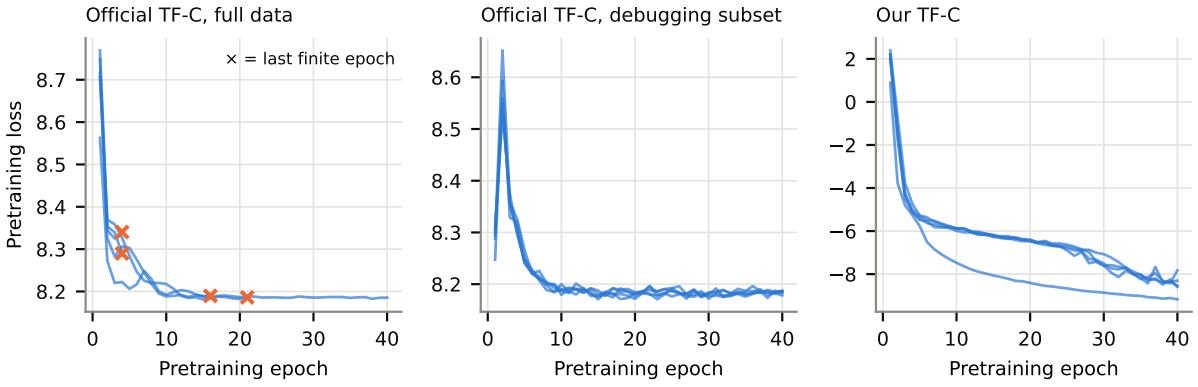


Figure 4: TF-C pretraining loss per epoch, one line per seed. Left: official code, full data (third attempt); × marks the last finite epoch. Middle: official code, debugging subset. Right: our TF-C; its loss is defined differently (D41 item 4), so the scales are not comparable.

5 Herning test rig

The Herning run `scope_20260901_112732` is a validation run of an unfinished rig (D7, D22): one bearing, one lubricant, rpm swept up and down at 13 temperature targets. The regression target is κ, computed from rpm and temperature as in the MSSP code (D55). Figure 5 shows the operating points.

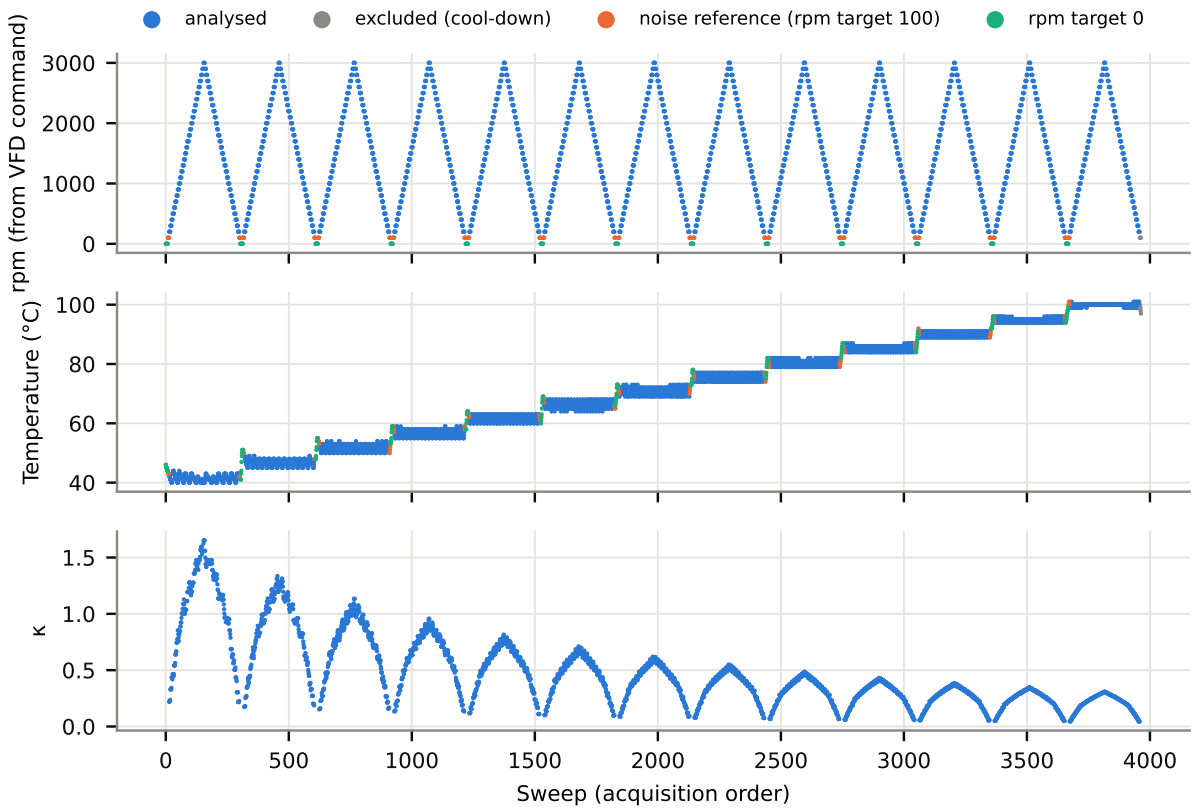


Figure 5: Herning run: rpm (from the frequency-converter command), temperature and κ for every sweep, in acquisition order, coloured by role in the band check (§5.1; D52, D55).

5.1 Herning band check: where AE and US carry information

Date	2026-10-09
Decisions	D51, D52, D55; led to D60, D61, D62, Q3
Runs	`20261009_085307_band_check_herning`

Question. Before band-limiting and downsampling the rig signals (D51): in which frequency bands do AE and US carry signal from the rotating bearing, and does that signal depend on the operating point (D52)?

Test. `eda/herning/check_bands.py` (then in `scripts/`; moved by D68) on the source file. Per sweep and sensor: Welch power spectrum (Hann window, 65,536 samples, 50 % overlap; 38-Hz resolution at 2.5 MHz), integrated over a low band (0–891 Hz) and base-10 third-octave bands from 1 kHz to Nyquist. Sweeps: 3,964 in total; 3,704 analysed (rotating, D55 exclusions applied), 126 noise reference (rpm target 100: bearing not rotating), 129 at rpm target 0, 5 excluded (cool-down). Per band: ratio of the median power of analysed sweeps to the noise reference (dB), and

Spearman correlation of band power with rpm, temperature and κ (κ from 0.045 to 1.651). Extra control against slow drift over the 13-hour run: the correlation with rpm computed within each temperature block (median over blocks). Run in four chunks of sweeps (3-minute limit per call), merged into one result (`NOTE.txt`). An earlier folder (`20261009_085150_band_check_herning`) contains only the config.

Result. Figure 6; all bands in Table 10 (Appendix B).

- AE: power above the reference from about 12.6 kHz to about 400 kHz, with peaks of 24.4 dB at 25.12 kHz and 24.0 dB at 100 kHz. In the bands from 12.6 to 398 kHz the within-block correlation with rpm is 0.96–0.99. From 631 kHz up there is no excess power (at most −0.02 dB) and band power *falls* with rpm within blocks (ρ from −0.72 to −0.64).
- US: 35–49 dB above the reference up to 6.3 kHz. Between about 40 and 200 kHz the excess is only 0.6–2.6 dB, but band power still rises with rpm within blocks (ρ 0.62–0.85) and correlates with κ (ρ 0.53–0.81).

Finding. AE shows rpm-dependent power above the reference roughly between 12 and 560 kHz (the origin of the part below 50 kHz is unknown, D60); above about 560 kHz it shows no excess power, and the user regards it as noise (D61, D62). US carries most of its power below about 8 kHz, but rpm-dependent content continues up to about 200 kHz. Correlation within temperature blocks rules out slow drift, but not electrical interference that also scales with speed (Q3).

Consequence. Feature bands: AE 50–560 kHz (nothing below the 50-kHz coupler high-pass; D60, D61), US 0.71–178 kHz (D60). Downsampling: AE by 2 (1.25 MHz), US by 5 (500 kHz) (D62). Whether US above about 40 kHz is bearing signal or interference is open (Q3).

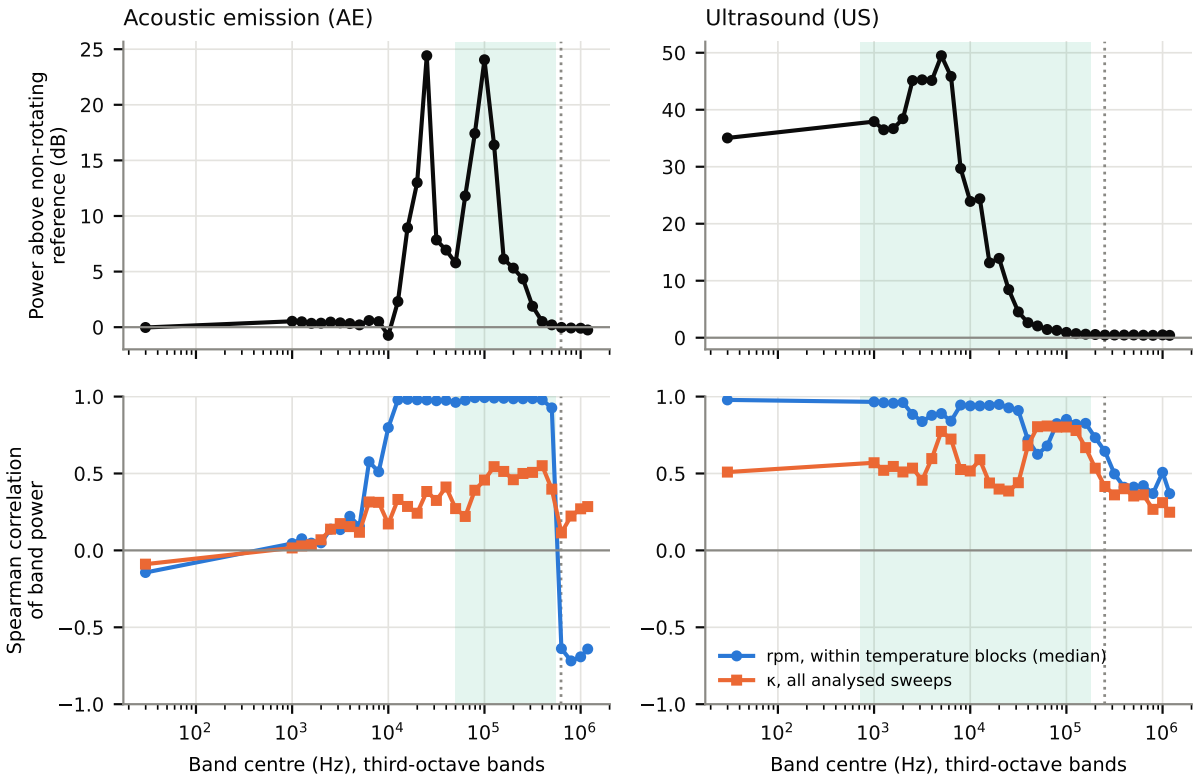


Figure 6: Herning band check. Top: median band power of rotating sweeps relative to the non-rotating reference. Bottom: Spearman correlation of band power with rpm (median within temperature blocks) and with κ (all analysed sweeps). Shaded: feature bands (D60, D61). Dotted: Nyquist frequency after downsampling (D62). The leftmost point is the 0–891-Hz band.

6 Open points

From this document:

- Data layer (§2.1): test outcomes are not stored as files.
- FordA (§3.2): TS-JEPA and PatchTST not yet in the comparison; the untrained-encoder control (D75 f) not run.
- Official TF-C, full data (§4.1): why seed 4 has no fine-tuning output.
- Our TF-C (§4.2): whether it counts as a reproduction, and whether the reproduction criterion needs a condition on the spread.
- FD-B (§4.4): what follows for the TF-C study (D35) from the FD-B diagnostics.

For supervision (`QUESTIONS.md`): Q1 Herning split (temperature hold-out details); Q2 comparing whole-sweep features with window-based SSL representations; Q3 US above 40 kHz and AE above 560 kHz — bearing signal or electrical interference.

A Run index

Table 9: Run index: every folder in `runs/` (D98). Commit with *: the working tree had uncommitted changes when the run started, so the commit alone does not reproduce the code.

Run folder (`runs/`)	What	Status	Commit	Device
`20261007_214220_ts2vec_forda_cpu_check`	TS2Vec on FordA, CPU check in the cloud workspace	Superseded by 20261008_084722_ts2vec_forda	none*	cpu
`20261008_084722_ts2vec_forda`	TS2Vec on FordA (reproduction)	Superseded by 20261020_144711_ts2vec_forda (rerun with timing)	adf16c6	cuda
`20261008_092321_tloss_forda`	T-Loss on FordA (reproduction)	Superseded by 20261020_144932_tloss_forda (rerun with timing)	5320251*	cuda
`20261008_134951_tfc_official_fd`	Official TF-C, full FD-A data, attempt 1	In §4.1	87e9222*	cuda
`20261008_142751_tfc_fd`	Our TF-C, FD-A to FD-B	In §4.2	87e9222*	cuda
`20261008_143021_tfc_official_fd`	Official TF-C, full FD-A data, attempt 2	In §4.1	87e9222*	cuda
`20261008_151014_probe_tfc_fd`	Frozen probes on our TF-C encoders	In §4.3	87e9222*	cuda
`20261008_155516_tfc_official_fd`	Official TF-C, full FD-A data, attempt 3	In §4.1	c11957a*	cuda
`20261008_162447_probe_spectrum_fd`	Probes on the |FFT| spectra of FD-B	In §4.4	none	–
`20261008_171159_tfc_official_fd_subset`	Official TF-C, debugging subset	In §4.5	c11957a*	cuda
`20261009_085150_band_check_herning`	Herning band check, stopped before writing results	Empty	c11957a*	–
`20261009_085307_band_check_herning`	Herning band check	In §5.1	c11957a*	–
`20261012_205347_spectra_herning`	Herning spectra	Examination (QUESTIONS.md, Q3)	5dba814*	–
`20261012_210217_vfd_lines_herning`	Frequency-converter lines in the Herning spectra	Examination (QUESTIONS.md, Q3)	5dba814*	–
`20261014_herning_examination`	Herning dataset examination	Examination (eda/herning)	–	–

Run folder (`runs/`)	What	Status	Commit	Device
`20261016_114600_tsjepa_forda`	TS-JEPA on FordA (official masking)	In §3.1 only; not in the comparison (D102)	5bde87a*	cuda
`20261016_114616_tsjepa_forda_fixed`	TS-JEPA (masking fixed) on FordA	In §3.2	5bde87a*	cuda
`20261017_115419_collapse_forda`	Collapse check on FordA	Not written up yet	7a69724*	cuda
`20261019_130910_patchtst_forda`	PatchTST on FordA (last-patch pooling)	Superseded by 20261019_154248_patchtst_forda_mean (same models, D87)	7a69724*	cuda
`20261019_154248_patchtst_forda_mean`	PatchTST re-encoded with mean pooling (D86, D87)	In §3.2	none*	cpu
`20261019_154305_patchtst_forda_flatten`	PatchTST re-encoded with flatten pooling (D86)	Not written up yet	none*	cpu
`20261020_135846_ablation_forda_A`	TF-C ablation on FordA, step A	Not written up yet	beee5e8*	cuda
`20261020_140639_ablation_forda_B`	TF-C ablation on FordA, step B	Not written up yet	beee5e8*	cuda
`20261020_141430_ablation_forda_C`	TF-C ablation on FordA, step C	Not written up yet	beee5e8*	cuda
`20261020_143928_raw_forda`	Baseline: raw samples (and floor) on FordA	In §3.2	beee5e8*	cpu
`20261020_144122_fft_forda`	Baseline: |FFT| on FordA	In §3.2	beee5e8*	cpu
`20261020_144235_catch22_forda`	Baseline: catch24 on FordA	In §3.2	beee5e8*	cpu
`20261020_144711_ts2vec_forda`	TS2Vec on FordA, with timing	In §3.1, §3.2	beee5e8*	cuda
`20261020_144932_tloss_forda`	T-Loss on FordA, with timing	In §3.1, §3.2	beee5e8*	cuda
`20261020_152809_minirocket_forda`	Baseline: MiniRocket on FordA	In §3.1, §3.2	beee5e8*	cpu
`20261021_104231_ablation_forda_D`	–	Not indexed	40618b6	cuda
`20261021_114216_ablation_forda_E`	–	Not indexed	f9bc9eb*	cuda
`20261021_132513_ablation_forda_F`	–	Not indexed	92e3c85*	cuda
`20261021_134511_ablation_forda_G`	–	Not indexed	92e3c85*	cuda
`20261021_140653_ablation_forda_H`	–	Not indexed	92e3c85*	cuda

B Herning band check: all bands

Table 10: Herning band check, all bands. Ratio: median band power of analysed sweeps over that of the non-rotating reference. ρ: Spearman correlation of band power with rpm (all analysed sweeps, and median within temperature blocks), temperature and κ.

Sensor	Band	Ratio (dB)	ρ rpm	ρ rpm in block	ρ temp.	ρ κ
AE	0-891.25 Hz	−0.0	−0.08	−0.14	0.13	−0.09
AE	1 kHz	0.5	0.03	0.04	0.10	0.02
AE	1.259 kHz	0.5	0.06	0.08	0.12	0.03
AE	1.585 kHz	0.3	0.04	0.05	0.08	0.04
AE	1.995 kHz	0.4	0.07	0.05	0.07	0.07

Sensor	Band	Ratio (dB)	ρ rpm	ρ rpm in block	ρ temp.	ρ κ
AE	2.512 kHz	0.5	0.16	0.14	0.04	0.14
AE	3.162 kHz	0.4	0.16	0.13	−0.01	0.17
AE	3.981 kHz	0.3	0.20	0.22	0.04	0.15
AE	5.012 kHz	0.2	0.08	0.16	0.00	0.12
AE	6.31 kHz	0.6	0.55	0.58	0.09	0.31
AE	7.943 kHz	0.5	0.48	0.51	0.05	0.31
AE	10 kHz	−0.7	0.70	0.80	0.30	0.17
AE	12.59 kHz	2.3	0.90	0.98	0.29	0.33
AE	15.85 kHz	8.9	0.88	0.98	0.35	0.29
AE	19.95 kHz	13.0	0.86	0.98	0.40	0.24
AE	25.12 kHz	24.4	0.90	0.98	0.24	0.38
AE	31.62 kHz	7.8	0.87	0.97	0.29	0.32
AE	39.81 kHz	6.9	0.91	0.98	0.20	0.41
AE	50.12 kHz	5.8	0.86	0.96	0.35	0.27
AE	63.1 kHz	11.8	0.85	0.98	0.42	0.22
AE	79.43 kHz	17.4	0.95	0.99	0.25	0.39
AE	100 kHz	24.0	0.96	0.99	0.17	0.46
AE	125.9 kHz	16.4	0.98	0.99	0.07	0.54
AE	158.5 kHz	6.1	0.97	0.99	0.11	0.51
AE	199.5 kHz	5.3	0.96	0.99	0.17	0.46
AE	251.2 kHz	4.3	0.97	0.99	0.12	0.50
AE	316.2 kHz	1.9	0.96	0.99	0.11	0.51
AE	398.1 kHz	0.5	0.97	0.98	0.05	0.55
AE	501.2 kHz	0.2	0.83	0.93	0.16	0.40
AE	631 kHz	−0.0	−0.43	−0.64	−0.34	0.11
AE	794.3 kHz	−0.1	−0.33	−0.72	−0.46	0.22
AE	1000 kHz	−0.1	−0.27	−0.69	−0.49	0.27
AE	1259 kHz	−0.2	−0.25	−0.64	−0.50	0.28
US	0-891.25 Hz	35.0	0.96	0.98	0.09	0.51
US	1 kHz	37.9	0.96	0.97	0.02	0.57
US	1.259 kHz	36.5	0.93	0.96	0.07	0.52
US	1.585 kHz	36.7	0.94	0.96	0.04	0.55
US	1.995 kHz	38.4	0.95	0.96	0.09	0.51
US	2.512 kHz	45.1	0.85	0.88	−0.00	0.53
US	3.162 kHz	45.3	0.82	0.84	0.07	0.46
US	3.981 kHz	45.1	0.84	0.88	−0.09	0.60
US	5.012 kHz	49.5	0.81	0.89	−0.31	0.77
US	6.31 kHz	45.8	0.78	0.84	−0.26	0.72
US	7.943 kHz	29.7	0.91	0.94	0.06	0.53
US	10 kHz	23.9	0.91	0.94	0.06	0.52
US	12.59 kHz	24.4	0.90	0.94	−0.03	0.59
US	15.85 kHz	13.1	0.88	0.94	0.15	0.44
US	19.95 kHz	13.9	0.88	0.95	0.21	0.40
US	25.12 kHz	8.4	0.85	0.93	0.21	0.39
US	31.62 kHz	4.5	0.85	0.91	0.14	0.44
US	39.81 kHz	2.6	0.64	0.72	−0.25	0.68
US	50.12 kHz	2.1	0.60	0.62	−0.42	0.80
US	63.1 kHz	1.4	0.63	0.68	−0.42	0.81
US	79.43 kHz	1.3	0.76	0.82	−0.35	0.80
US	100 kHz	0.9	0.78	0.85	−0.35	0.80
US	125.9 kHz	0.7	0.74	0.82	−0.36	0.78
US	158.5 kHz	0.6	0.79	0.83	−0.21	0.67
US	199.5 kHz	0.6	0.71	0.73	−0.11	0.53
US	251.2 kHz	0.5	0.64	0.64	−0.01	0.42
US	316.2 kHz	0.5	0.53	0.50	−0.03	0.36
US	398.1 kHz	0.5	0.40	0.41	−0.15	0.40
US	501.2 kHz	0.5	0.37	0.41	−0.11	0.35

Sensor	Band	Ratio (dB)	ρ rpm	ρ rpm in block	ρ temp.	ρ κ
US	631 kHz	0.4	0.40	0.42	−0.10	0.36
US	794.3 kHz	0.4	0.37	0.37	−0.01	0.27
US	1000 kHz	0.5	0.51	0.51	0.04	0.31
US	1259 kHz	0.4	0.38	0.37	0.01	0.25